

CCTV-Based Digital Twin System

System Modeling and Functional Requirements

1 System Overview

The proposed system is a CCTV-based Digital Twin Platform designed to monitor people through camera feeds, identify and track individuals, and generate continuously updated digital profiles (digital twins).

The system combines:

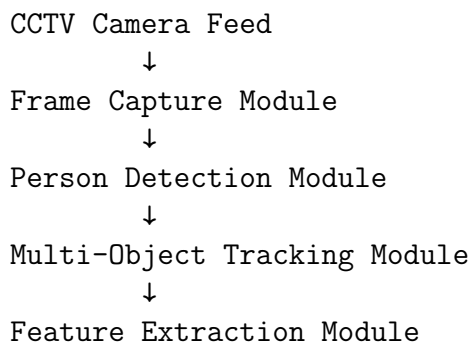
- Computer Vision
- Artificial Intelligence
- Real-Time Tracking
- Data Storage
- Identity Management

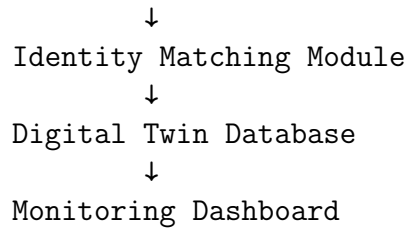
The primary objective is to create a system capable of:

- detecting people from CCTV feeds,
- maintaining persistent identities,
- storing behavioral and visual data,
- and updating digital profiles over time.

2 System Architecture Model

2.1 High-Level Architecture





3 Core System Modules

3.1 Frame Capture Module

Purpose: Captures real-time video streams from CCTV cameras.

Inputs:

- RTSP stream / Webcam feed / Video feed

Outputs:

- Individual video frames

Responsibilities:

- Continuous frame acquisition
- Frame preprocessing
- Resolution adjustment
- Frame buffering

3.2 Person Detection Module

Purpose: Detects humans within video frames.

Technology:

- YOLOv8 object detection model

Inputs:

- Video frames

Outputs:

- Bounding boxes of detected persons

Responsibilities:

- Human detection
- Confidence scoring
- Coordinate generation

3.3 Multi-Object Tracking Module

Purpose: Tracks detected individuals across consecutive frames.

Technology:

- DeepSORT / ByteTrack

Inputs:

- Detection bounding boxes

Outputs:

- Temporary tracking IDs

Responsibilities:

- Assign IDs to detected persons
- Maintain identity consistency across frames
- Handle movement tracking

3.4 Feature Extraction Module

Purpose: Extracts identifiable features from detected individuals.

Inputs:

- Person image crops

Outputs:

- Feature vectors / embeddings

Features Extracted:

- Face embeddings
- Clothing color
- Height estimation
- Movement patterns

Technologies:

- DeepFace
- FaceNet
- InsightFace

3.5 Identity Matching Module

Purpose: Determines whether a detected individual already exists in the database.

Inputs:

- Feature embeddings
- Existing database embeddings

Outputs:

- Existing Person ID
- OR
- New Person ID

Responsibilities:

- Similarity comparison
- Identity persistence
- Duplicate prevention

3.6 Digital Twin Database Module

Purpose: Stores digital profiles and historical data.

Database Options:

- MongoDB
- PostgreSQL

Stored Data:

- Person ID
- Face embeddings
- Snapshots
- Visit history
- Movement trajectory
- Behavioral statistics

3.7 Monitoring Dashboard

Purpose: Provides visualization and monitoring interface.

Features:

- Live camera feed
- Person tracking visualization
- Profile viewing
- Search functionality
- Activity timeline

4 Functional Requirements

4.1 FR-1: Video Stream Acquisition

Description: The system shall capture video streams from CCTV cameras continuously.

Inputs:

- Live CCTV stream

Outputs:

- Processable frames

4.2 FR-2: Human Detection

Description: The system shall detect human presence in video frames.

Expected Behavior:

- Detect multiple persons simultaneously
- Generate bounding boxes

4.3 FR-3: Person Tracking

Description: The system shall maintain consistent tracking IDs for detected persons across frames.

Expected Behavior:

- Same person retains same tracking ID during movement

4.4 FR-4: Feature Extraction

Description: The system shall extract identifiable visual features from detected individuals.

Features:

- Face embeddings
- Clothing appearance
- Spatial position

4.5 FR-5: Identity Matching

Description: The system shall compare extracted features with stored profiles to identify repeated individuals.

Expected Behavior:

- Match returning individuals
- Create new profiles for unknown individuals

4.6 FR-6: Digital Profile Creation

Description: The system shall create a digital twin profile for each unique person.

Stored Information:

- Unique ID
- Detection timestamps
- Appearance snapshots
- Historical movement data

4.7 FR-7: Historical Data Management

Description: The system shall maintain historical records for future analysis.

Data Stored:

- Entry/exit time
- Visit frequency
- Camera locations

4.8 FR-8: Real-Time Processing

Description: The system shall process incoming frames with minimal delay.

Requirement:

- Near real-time performance

4.9 FR-9: Multi-Person Handling

Description: The system shall support simultaneous tracking of multiple individuals.

4.10 FR-10: Data Retrieval

Description: The system shall allow retrieval of digital twin profiles using unique person IDs.

5 Non-Functional Requirements

5.1 Performance

- Real-time processing capability
- Low latency detection and tracking

5.2 Scalability

- Support multiple cameras
- Handle large profile databases

5.3 Reliability

- Continuous system operation
- Stable identity tracking

5.4 Security

- Secure storage of biometric information
- Access-controlled database

5.5 Privacy

- Minimize unnecessary personal data storage
- Ensure ethical data handling

6 Initial Prototype Scope

The first prototype will focus only on:

- live video capture,
- human detection,
- tracking,
- snapshot storage,
- and simple digital profile generation.

Advanced features such as:

- behavioral prediction,
- emotion analysis,
- and multi-camera synchronization

are outside the initial implementation scope.

7 Proposed Technology Stack

Component	Technology
Programming Language	Python
Computer Vision	OpenCV
Object Detection	YOLOv8
Tracking	DeepSORT
Face Recognition	DeepFace / InsightFace

Database	MongoDB
Backend API	FastAPI
Frontend Dashboard	React / HTML

8 Expected Outcome

The final system should be capable of:

- detecting people from CCTV feeds,
- maintaining persistent identities,
- generating digital profiles,
- and updating those profiles continuously using real-time observations.